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Identifying greenwashing in corporate-social responsibility reports using natural-language processing

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Abstract

A textual analysis of corporate-social responsibility (CSR) reports reveals that companies engaged in environmental violations report differently from firms with a clean record. The violators issue longer, more positive and more frequent reports to relay environmental content that is more copious but less readable. The violator firms appear to modify their reporting practices right after committing a violation. The findings suggest that culpable firms exploit the current unregulated–unaudited state of CSR reporting as a means of greenwashing and call for institutional change. Our results are robust to a number of industry-firm characteristics, including board composition, ownership dispersion and international presence.

KEYWORDS

corporate-social responsibility, greenwashing, natural-language processing, sustainability reporting

JEL CLASSIFICATION

G30, G32, M14

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1 | INTRODUCTION

Over the past decade, public apprehension regarding the state of the environment has compelled companies to pursue ecofriendly policies. By addressing citizens' ethical concerns, companies are believed to extract a number of benefits, including customer loyalty, increased sales, lower cost of capital and enhanced employee retention.

However, a strategy that focuses on achieving beneficial outcomes for the environment, in addition to financial gains, could be costly and ultimately unsuccessful in maximizing shareholder value. This dichotomy of incentives has led to the phenomenon of "greenwashing", the practice by which corporations make spurious claims with respect to environmental actions in an effort to rip the benefits of an environmentally conscious image without incurring the costs associated with environmental practices. How widespread is greenwashing, and what are the channels of its dissemination? In this study, we show that corporate-social responsibility (CSR) reports, in their current unregulated and unaudited form, are exploited by culpable firms as a means of greenwashing. We call for greater regulation oversight of CSR reporting, including effective mechanisms for assurance.

Identifying greenwashing involves comparing environmental claims versus actual environmental performance. Both sides of the equation are hard to measure. Firms broadcast environmental claims in different formats and through diverse means of communication, which makes the aggregation and overall evaluation of such information difficult. At the same time, quantifying environmental performance, particularly on a short-term basis, is inherently ambiguous, leading to a wide divergence of experts' opinions on a firm's environmental record. We address the issues of measurability first by focusing on a single source of information, the CSR report, as the *primoris* (foremost) means of corporate communication, typically signed by the company's chief executive officer (CEO) or responsible officer. Likewise, we seek a reliable "yardstick" of actual environmental performance to juxtapose against environmental assertions. We opt for a simple, yet error-free, binary measure of environmental responsibility, that is, whether a firm has received a monetary penalty for environmental violations or not. This process allows us to classify firms to violators versus nonviolators and directly examine variation in the environmental claims between the two groups. To formulate testable hypotheses that are economically meaningful, we rely on theories of greenwashing via decoupling, selective disclosure and attention deflection.

Our investigation has two objectives: first, to assess whether the CSR reporting practices of "green" firms and "brown" firms differ, and second, to investigate whether a company changes reporting practices straight after committing a violation. Our empirical analysis focuses on a sample of US public firms for the period 2008–2022, a time in which the number of American firms releasing CSR reports rose exponentially,[1] in line with public environmental concerns. Given the voluntary, unaudited and unstructured nature of CSR reporting in the United States over this period, to what end were these reports issued, and what can they tell us about the future? We find that violator firms tend to issue longer, more positive and more frequently available reports and use these reports to broadcast stronger, but less readable, environmental content. Issuing longer reports and maintaining these reports on the public domain for longer,

[1]S&P 500 firms that released sustainability or corporate responsibility reports (CSR) between 2011 and 2019 have risen from 20% to 86% over the period. https://www.ga-institute.com/press-releases/article/flash-report-86-of-sp-500-indexR-companies-publish-sustainability-responsibility-reports-.

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suggests that violator firms are willing to divulge more information in line with selective disclosure theory. Populating the report with greater amounts of environmental content, albeit with a lower degree of readability, likewise points out to attention deflection and decoupling. Firms appear to modify their CSR reports in this direction following the year of environmental violation. From the onset, we control for firm/industry-specific differences using a propensity-score methodology (PSM) that generates a matched sample. We thereon extend our analysis to include known characteristics of CSR performance, such as board composition, ownership dispersion and international presence. Overall, we provide evidence in support of the idea that firms with poor environmental records exploit the present lack of regulatory oversight to channel false environmental claims through CSR reports.

The contributions of the paper are manifold. Using standard natural-language processing (NLP) techniques, we analyze the textual content of CSR reports to quantify their environmental content, tone (sentiment) and readability. On the basis of a pre-established Lexicon of environmental terms (DiCoEnviro), designed by linguists for NLP, we employ a dictionary approach to extract, for the first time to our knowledge, an environmental score (ES) for the textual content of unstructured CSR reports. We thereon interact the ES with each of the readability and positiveness scores, allowing us to test directly for greenwashing via decoupling and attention deflection. Looking at the size and reporting frequency of the reports, we also examine greenwashing via selective disclosure. To "cut through the noise" of environmental, social, and governance (ESG) third-party ratings, we opt for a binary, error-free proxy for environmental performance: whether a company committed a penalized environmental violation or not. We thus suggest a stricter criterion for disentangling "brown" firms from "green" firms, one which provides positive assurance for violators and/but only negative assurance for nonviolators. This criterion is used to assess the overall differences in CSR reporting between violator and nonviolator firms in univariate and multivariate tests, as well as to detect within-firm changes in CSR reporting following a transition to violator using cosine similarity and staggered difference-in-differences (DiDs) regressions.

Our study lies in the cross-section of management, corporate finance and accounting literature for sustainability reporting. More so, the paper is linked to, but remains distinct from, a relatively recent set of empirical studies that focus on the textual analysis of CSR reports to detect greenwashing. Michelon et al. (2015) examine the quality of CSR disclosure for a small sample of London Stock Exchange-listed firms between 2005 and 2007, using content analysis and manual coding. Lang and Stice-Lawrence (2015) use natural-language techniques to analyze reporting data, but their study focuses on annual reports. Clarkson et al. (2020) provide a large-scale textual analysis of CSR reports, but do not examine greenwashing. Albitar et al. (2023) likewise apply NLP to sustainability sections of annual reports, but their analysis is limited to detecting tone.

Our results have important implications for research and industry. As the practice of issuing stand-alone CSR reports and annual reports becomes established, we highlight the discrepancy between the two kinds of reports: the former is voluntary, unstructured and unaudited; the latter is statutory and subject to independent assurance.

Closing the regulation gap between financial and sustainability reporting is necessary to ensure that CSR reports are held to the same standards of accountability as annual reports. Previous efforts to regulate CSR reporting more strongly, as evidenced by European and worldwide initiatives, have been of limited success. Haji et al. (2023) find that "CSR reporting continues to be ceremonial rather than substantive after the regulations". Christensen et al. (2021) caution on potential actual costs of enforcing mandated disclosures for CSR, such as loss

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of competitiveness, exposure to litigation risk and resorting to greenwashing. These concerns are somehow alleviated by Fiechter et al. (2022), who find that the real effects of the European Union's CSR directive have been beneficial, leading to meaningful CSR activities and actual CSR outcomes. Given our present findings of greenwashing in CSR reports, we are more inclined to support the idea of a statutory CSR reporting and auditing framework for US firms despite the anticipated adjustment costs.

The rest of the paper is organized as follows. Section 2 presents the literature review. Section 3 develops testable hypotheses. Section 4 outlines the research design. Section 5 provides empirical results, while Section 6 concludes.

2 | LITERATURE REVIEW

2.1 | Societal expectations

Corporations are keen to promote an environmentally conscious image. Capitalizing on widespread concern for the state of the environment, companies that appear to be good corporate citizens enjoy preferential investment opportunities (Sen et al., 2006), lower financial risk (Orlitzky & Benjamin, 2001), increased employee retention (Greening & Turban, 2000; Turban & Greening, 1997) and greater customer loyalty (Martin et al., 2009). At the same time, continuous demand for ecofriendly assets has led to a proliferation of "green funds". According to the US SIF Foundation, approximately $8.4 trillion (SIF Annual Report, December 2023) in US-domiciled assets, or about 12.5% of total US assets, are deployed using sustainable investing strategies.

Over the last 10 years, ecofriendly investments have yielded superior returns (Edmans, 2011; Nagy et al., 2016). Pástor et al. (2022) show that the higher-than-expected returns on green assets are not in line with ex-ante estimates of required returns. Instead, they are more likely to reflect a willingness to pay an additional price—a "greenium", out of environmental considerations. Baker et al. (2022) find that on average, investors are willing to pay 20 basis points more to a fund with an ESG mandate than to an otherwise identical investment. Consumers also appear willing to accommodate higher prices for environmentally friendly products, ranging from restaurant food (Namkung & Jang, 2017) to air travel (Hinnen et al., 2017) and anything in between.

As public sentiment concerning the state of the environment continues to drive preferences, more companies attempt to harvest the "greenium" by broadcasting ecofriendly policies. The information is only sometimes accurate. A report released by the European Commission as part of its 2021 annual screening of websites ("sweep") found that 42% of the examined environmental claims were likely "exaggerated, false or deceptive". In a similar exercise performed globally, the International Consumer Protection Enforcement Network reported that 40% of the sampled websites were likely using misleading tactics to appear ecofriendlier. On November 22, 2022, Amundi, Europe's largest investment manager, downgraded most of its €45bn ($47bn) of funds ranked Article 9, the highest grade in the EU's sustainability disclosures, to Article 8, or "light green", suggesting the funds were misclassified in the first place. The practice by which companies appear to make spurious claims concerning environmental actions—"greenwashing" is the subject of research that it is both extensive and multidisciplinary. While comprehensive reviews can be found in De Freitas Netto et al.

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(2020) and Yang et al. (2020), we summarize some key findings of the literature and, importantly, the limitations that prompt further investigation.

2.2 | Greenwashing

Greenwashing is multifaceted, each dimension presenting a separate identification, measurement and inference challenge. A common type of greenwashing arises through selective disclosure of: "positive information about a company's environmental or social performance, while withholding negative information on these dimensions", so as to create an overly positive corporate image (Lyon & Maxwell, 2011, p. 5). In selective disclosure, the act of greenwashing relates to how information is classified and disseminated (Font et al., 2012; Yang et al., 2020). An important implication of the selective disclosure practice is that companies with poor environmental records are likely to divulge more information (Patten, 2002) than their green peers.

A second form of greenwashing refers to decoupling, the gap between symbolic and substantive corporate-social actions (Walker & Wan, 2012). Unlike selective disclosure, decoupling is associated with corporate claims and actions, enabling companies with more negative CSR performance to project more positive communication. Attention deflection activities to conceal poor environmental performance, or outright deceptive manipulation (i.e., Siano et al., 2017), including the use of dubious ecofriendly labels and rogue third-party authorizers (De Freitas Netto et al., 2020) are common tools of decoupling. A final distinction in different types of greenwashing rests on how the message of ecofriendliness is delivered: Claim greenwashing uses textual arguments that explicitly advertise environmentally friendly products or corporate policies, while executional greenwashing (Parguel et al., 2015) uses nature-evoking images (i.e., wind turbines, green/blue images, natural landscapes, etc.) that may be irrelevant to actual products or services.

While firm incentives for being perceived as ecofriendly are numerous, the motivation to actively mislead is more complex. An incentive to cheat lies in that the pursuit of environmental or other societal goals may result in reduced profitability (Bénabou & Tirole, 2010), either by allowing suboptimal cost control or truncating the set of available investments should those not meet extended criteria. Furthermore, deviations from the management's ingrained mission to maximize shareholder value raise crucial corporate governance issues (Cespa & Cestone, 2007) that may come at a sacrifice for the shareholders. Menla Ali et al. (2023) highlight that firms may resort to "greenwashing" when the career stability of powerful CEOs and the corporate control exerted by major shareholders are at risk.

On the other hand, overcommitting to stated environmental goals and failing to deliver is costly, as the discrepancy between CSR performance and reporting draws consumer backlash and punishment by civil organizations (Lyon & Maxwell, 2011). Ioannou et al. (2022) show that perceived "greenwashing" is understood as a form of corporate hypocrisy and is negatively associated with customer satisfaction. Discovered greenwashing has an asymmetric impact in reducing corporate legitimacy (Guo et al., 2018) compared with verifiable activities that increase corporate legitimacy.

Hence, with growing scrutiny on environmental claims, it remains a subject of empirical investigation, whether companies engage in greenwashing as a profit-maximizing strategy, or are driven by nonpecuniary motives of shareholders (Hong & Shore, 2023).

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2.3 | CSR disclosure and environmental performance

This line of inquiry, relating advertised CSR actions with realized CSR performance, presents significant challenges to regulators and researchers. Both sides of the equation appear to be noisy, and efforts to approximate the two variables for empirical association tests are incomplete. The first issue relates to how one discerns and aggregates what is claimed. CSR information is released through various communication channels, including, but not limited to, news feeds on the corporate websites, stakeholder engagement via social media, press releases, advertising, product packaging, becoming a signatory to responsible investment frameworks[2] and issuing stand-alone CSR reports. Each outlet is associated with varying degrees of accountability, scope, format and target audience. The fragmentation in CSR reporting makes it difficult to assess the overall quality and reliability of management's environmental assertions. Furthermore, due to the heterogeneity of the information signals, it is hard to aggregate the information to construct a comprehensive picture of environmental claims.

Singling out the stand-alone CSR report as a more formal means of communication does not alleviate heterogeneity. Unlike traditional financial reporting, CSR reports are not mandated by the Securities and Exchange Commission (SEC, 2020) and are not prepared according to Generally Accepted Accounting Principles. As such, CSR reports differ substantially in the diversity of topics, objective functions, time horizon and, ultimately, users and uses. The amount of CSR disclosure, which remains voluntary, also varies across firm characteristics, such as industry membership, size and ownership dispersion. Given substantial firms' incentives to voluntarily disclose CSR/ESG information, a lack of statutory framework does not so much hinder the amount of information released as much as compromises the quality and credibility of management's assertions.

A second point of contention arises from the difficulty of measuring actual CSR performance. This is an issue arising from diverse and incomplete auditing. In the absence of SEC regulations and in line with fragmented reporting, a miscellaneous set of institutions appear to have taken the role of quasi-watchdogs. At times, these include nongovernmental organizations, activist investors, consumer protection agencies and financial institutions that provide a third-party assessment of corporations' ESG performance. Currently, ESG rating providers, KLD, Sustainalytics, Moody's ESG, London stock exchange group (Refinitiv), S&P Global and MSCI amongst the most prominent, appear to be far-reaching in influencing investment decisions and corporate policies as a growing number of investors rely on ESG ratings to form portfolio choices.

Nevertheless, these agencies appear to provide a widely divergent test of performance. Berg et al. (2022) find that ESG ratings from the six most prominent rating agencies correlate by as little as 38%, and differ in scope, weighting and measurement. By comparison, the correlation of alternative credit ratings is 99%. This level of discord invalidates third-party ratings and highlights the difficulty of environmental auditing: If experts cannot agree on two out of three ESG ratings, who is to cry "greenwashing", and to what end? Larcker et al. (2022) doubt that "we agree on the purpose of ESG" as the number one "myth" in their list of ESG grievances.

[2]More than 3000 institutional investors and service providers have signed onto the Principles of Responsible Investment (2020), a commitment to integrate ESG/CSR issues into their investment analysis and decision-making processes.

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A closer look at ESG rating divergences attributes discord to the difficulty of agreeing on what is essential to measure and the challenge of quantifying largely qualitative indicators. Chatterji et al. (2016) establish two preconditions for rating convergence. The first one, called "theorization", refers to what precisely is rated and how important it is in the composition of an overall ESG score. Hence, the theory corresponds to the scope of assessment and the weight assigned to specific ESG markers. The difficulty of adopting a single system of taxonomy concerning what categories of ESG actions are considered and how much each action is valued could be traced to differences of opinion amongst the raters and business considerations concerning product differentiation and competitiveness. The second precondition, called "commensurability", refers to the actual measurement of CSR as defined in theorization. While divergence in theorization can be alleviated by reclassifying different categories preferred by raters, the measurement problem is more complex to circumvent. This is because raters are called upon to quantify inherently qualitative indicators, such as societal externalities from an improved environment, that are only tangentially related to the core business operations. In Berg et al. (2022), 56% of ESG rating divergence is attributed to measurement, 38% to scope and 6% to weight.

The paper contributes to the ongoing debate on corporate greenwashing by circumventing the problems of noisy signaling, complex theorization and low commensurability that hinder the problem of identification. To achieve this, we employ a simple, error-free, binary measure of environmental responsibility: whether the firm has received a monetary penalty for environmental violation, or not. Compared with a composite, qualitative measure of environmental performance our proxy is reliable in the sense that it is not subject to rating divergence and can be verified objectively. While we obtain the status of a firm being a "violator" or a "non-violator" from KLD/MSCI (2008–2019) and Refinitiv (2019–2022), the environmental transgressions linked to monetary penalties are factual and not subjective to the specific raters. This is in line with Berg et al.'s (2022, p. 1342) suggestion that "... researchers can construct hypotheses around more specific sub-categories of ESG performance, such as GHG Emissions or Labor Practices. This avoids the problems of weight and scope divergence ...". Our proxy is not subject to measurement error, particularly when used as binary, without quantifying the size of penalties. Furthermore, we note that violator status is a conservative measure of environmental performance because it provides positive assurance for environmental offenders and only negative assurance for nonoffenders. This distinction provides a stiffer "yardstick" for disentangling "brown" firms from "green" firms and a more reliable one, as it counters the optimistic bias of management assertions and the difficulty of measuring the actual realization of these assertions.

In turn, we contrast the violator status of the firm directly to the information content of its stand-alone CSR report. By focusing on a single source of information, we avoid the difficulty of aggregating information signals from diverse sources (i.e., corporate websites, social media, product labels, etc.). At the same time, we view the stand-alone CSR report as the most formal means of corporate communication and one that holds the most significant amount of accountability. Typically, the CSR report is signed by the company's CEO or designated CSR officer assuming responsibility for the content of the report. CSR reports are also maintained in public view longer than other communication media, suggesting that the company remains publicly liable to environmental claims for longer periods of time. Despite these benefits, the CSR report remains an unregulated, unstructured document that differs in scope, length, language and objectives. To extract the environmental content of the reports, at scale, we employ increasingly popular (Aziz et al., 2022) NLP techniques that facilitate the "reading" of

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large swaths of text beyond the capacity of (reasonably applied) human labor. We then use standard machine learning (ML) algorithms to quantify the amount of environmental content, degree of readability, positiveness and length.

3 | HYPOTHESES

The variables extracted allow us to test for greenwashing through the following baseline hypotheses:

H1. The probability of being an Environmental Violator is positively associated with making more environmental claims.

Our first hypothesis tests for greenwashing by decoupling (Walker & Wan, 2012), capturing the gap between symbolic representations of CSR and substantive corporate actions. To test this hypothesis the actual environmental status of the firm is identified through the classification of the firm as a violator or not, while the symbolic representations of environmental claims are measured by the amount of environmental content identified in the CSR report.

H2. The probability of being an Environmental Violator is positively associated with releasing more obfuscated information.

The second hypothesis tests for greenwashing by attention deflection (Guo et al., 2018). We postulate that violator firms conceal their actual environmental performance by divulging a greater number of environmental claims, but these claims are less clearly articulated to avoid detection. As with H1, the actual environmental status of the firm is identified through the classification of the firm as a violator or not, while the degree of information obfuscation is measured by the degree of readability of the CSR report.

H3. The probability of being an Environmental Violator is positively associated with releasing more positive information.

While the CSR report is a firm-issued document, which, by definition, is positive, we look at cross-sectional variation in the degree of positiveness using sentiment analysis (SA). We argue that environmental violators attempt to conceal poor environmental performance by projecting a more positive image than their green peers.

H4. The probability of being an Environmental Violator is positively associated with voluntarily disclosing more information.

This hypothesis relates to greenwashing by selective disclosure (Yang et al., 2020) a direct implication of which is that companies with poor environmental performance are likely to volunteer more information (Patten, 2002). To this end, we relate the environmental violator status of a firm to the actual size of its CSR report, as measured by the number of pages and words. Taking a step back, we also observe the frequency of violator firms releasing a CSR report and making the report publicly available, and compare this frequency with that of nonviolator firms. We likewise conjecture that violator firms are more likely to produce

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publicly available CSR reports. Hypotheses H1–H4 are tested independently, with univariate association tests, and jointly in multivariate logistic regressions. We also test for interactive effects between environmental content and positiveness and environmental content and readability.

Furthermore, we relate the probability of being an environmental violator to corporate finance variables known to affect ESG/CSR performance. These include board size and diversity (Velte, 2022), the international aspect of operations (Cai et al., 2016) and ownership dispersion (Nofsinger et al., 2019). In particular, we test for effects of board composition via:

H5. The probability of being an Environmental Violator is negatively associated with board size, the percentage of women directors on the board, and the number of outside directors on the board.

Traditionally, board size has been used as a catch-all proxy of corporate governance quality. Nevertheless, size is not unconditional as it presents a trade-off between increased monitoring (reduced agency costs) and the cost of this monitoring (Boone et al., 2007). Looking at board composition, a score of papers (Cronqvist & Yu, 2017; McGuinness et al., 2017 among others) suggests that female representation on board positively affects CSR policies and ESG performance. Board independence, as captured by the number of outside directors, is also found to be associated with better corporate governance (Knyazeva et al., 2013).

To test for international aspect effects, we formulate the following hypothesis:

H6. The probability of being an Environmental Violator is negatively associated with the firm's international operations.

Firms with a significant percentage of income from foreign operations and tax paid in foreign countries are identified as having an international presence. These firms are hypothesized to be less likely to commit environmental violations in the United States. For example, Iliev and Roth (2023) find that exposure of US firms to foreign countries' sustainability reforms (which are often more substantial than those in the United States) has a positive effect in shaping local CSR policies and increasing the firm's ESG performance.

Finally, we relate the probability of being an environmental violator to ownership dispersion. Our final hypothesis states:

H7. The probability of being an Environmental Violator is negatively associated with ownership dispersion.

To obtain a simple measure of ownership dispersion, we group firms identified in KLD/MSCI as "Institutions Dominant" or "Indexed Stock" or "Mixed Ownership" or "Widely Held" as higher dispersion firms, and firms classified as "Founder Firm" or "Family Firm" or having a "Principal Shareholder" or a "Controlling Shareholder" as lower dispersion firms. Evidence concerning ownership characteristics and ESG performance is mixed. For example, Borghesi et al. (2014) find a negative association between institutional ownership and CSR performance, while Chava (2014), focusing only on ESs, finds the opposite. Drobetz et al. (2024) show that institutional investor ownership is negatively related to corporate environmental costs. Nevertheless, the result is not unconditional, and it appears to be driven by long-term foreign institutional investors from advanced economies. The same ambiguity is encountered when

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looking at closely owned firms. While El Ghoul et al. (2016) find that family firms have worse ESG performance, Abeysekera and Fernando (2020) conclude that: "Family firms do at least as well as non-family firms in alleviating environmental concerns".

Corporate finance variables remain of interest as stand-alone determinants of the probability of a firm being an environmental violator. Nevertheless, we consider these variables more informative in multivariate regressions, controlling for confounding factors in the relationship between environmental performance and environmental claims (H1–H4). This aligns with the view that board and other firm characteristics are only measurable proxies for latent internal governance that drive CSR communication and environmental actions (Gillan et al., 2021).

4 | RESEARCH DESIGN

4.1 | Sample selection procedure

Because the public issuance of a CSR report is voluntary, the availability of CSR reports, particularly when subject to retrospective collection, is widely divergent across firms/years. Focusing, a priori, on the CSR reports that are readily available generates a skewed sample, towards larger firms, with narrow industry membership, and close environmental outlook. To avoid a sample selection bias and to construct a sample that is not based on the availability of CSR reports, we impose the following procedure:

We begin by anchoring our sample on the MSCI ESG KLD Stats data set, which provides a fair representation of firm population across industries and years, and has the highest coverage among the six most prominent rating agencies. We select all firms with nonmissing data on negative environmental indicators, whereas a zero score indicates a nonviolator firm/year instance, while a nonzero score indicates a violator firm/year instance (the exact definitions are provided in Appendix A). Furthermore, we extend the MSCI ESG KLD Stats data set, which covers the period from 2008 to 2019, with Refinitiv data set for additional 3 years from 2020 to 2022.[3] True to our direct measurement of environmental performance, we identify violator firms with the variable "Environmental Fines" (ENERDP103). Thus, the timeframe of the study is from 2008 to 2022.

Second, we apply PSM to produce a matched sample of violator/nonviolator firms in the KLD/MSCI/Refinitiv sets controlling for firm and industry/year characteristics obtained from Compustat database. It is clarified that we run the PSM routine for *each year* separately and with *annual replacement*. This is beneficial for two reasons: First, it allows for a sufficient number of correct matches, since companies tend to transition between violator status over time. In particular, after consolidating KLD/MSCI with Refinitiv and observing a wide cross-section of firms over 15 years, we find only very few companies that are perennial violators.[4] While there is a large number of always-green firms in the sample (control group), the majority of the violator firms (treatment group) got fines only occasionally, making it challenging to find satisfactory matches without annual replacement. Second, matching with replacement minimizes the propensity-score distance between the matched comparison and the treatment

[3]We thank an anonymous referee for suggesting to extend the data set for another 3 years.
[4]Some publicly known examples include Monsanto.

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units and is conducive in terms of bias reduction (Dehejia & Wahba, 2002). The matched data set preserves the stratification of the sample which remains representative.

Third, for the matched sample, we perform a fairly exhaustive search to retrieve publicly available CSR reports, cross-checking four sources: companies' own webpages, Sustainability Reporting, Corporate Register and GRI databases. If any additional variables are missing for any firm/year instance, the observation is dropped from the sample. We do not substitute the dropped observations, not to deviate from the sample selection methodology. The matched sample controls, from the onset, for unwanted factors underlying violator status and CSR reporting practices. The matching methodology including the selection of covarying factors is explained below.

4.2 | Propensity-score matching variables

The PSM routine generates a set of pairs between violator/nonviolator firms that are similar in ways that may bias the association of outcome (probability of being a violator) and treatment (explanatory variables). By looking at pairs that are "otherwise identical" the effect of confounding factors is eliminated while the stratification of the sample concerning these factors is maintained. Below, we motivate the set of control variables. The first control is firm Size, defined as the market capitalization of the firm. While larger firms are thought to extend more CSR activities and be more transparent (Chae, 2005), Udayasankar (2008) documents a U-shape relationship in which medium-sized firms are the least motivated to pursue CSR outcomes. At the same time, larger and smaller firms that appear more active in sustainable policies do so with different economic incentives. Firm Size is also a well-known determinant of board characteristics, ownership dispersion and international presence. Our second control is firm Age, defined as the number of years between the year of incorporation and the time of measurement. Withisuphakorn and Jiraporn (2016) find that more mature firms invest significantly more in CSR, and the effect is most strongly applied to diversity and environmental awareness. The finding is also confirmed by D'Amato and Falivena (2020), who argue that smaller and younger companies need more financial resources, experience, or reputation to pursue environmental policies.

The third control is intangibility, defined as 1 minus the ratio of tangible assets over the total assets of the firm. Edmans (2023) provides an exciting perspective that ESG policies are, in fact, intangible assets themselves, in line with, that is, quality of human capital or corporate culture. In Jun et al. (2022), the relationship between ESG disclosure and intangible capital is characterized by an inverted S-shaped curve in which initial ESG outlays increase intangible capital. At higher levels, the marginal effect of increasing ESG investment eventually diminishes intangibles.

We employ the following three control variables to account for differences in financial performance, valuation and leverage constraints. Return on Equity, defined as net income over last year's equity, captures immediate profitability. The relationship between firm profitability and CSR is not unidirectional. On the one hand, a company that struggles financially may be unable to invest in sustainability policies that will not directly improve its bottom line. At the same time, companies that maintain some capacity for CSR initiatives may aim at environmental outcomes to improve their brand image and ultimately increase profits (Cornett et al., 2016). The ratio of market value (market capitalization) to book value of equity is a catch-all proxy for investor confidence, expectations for future profitability, growth opportunities and

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cost of capital. Conventional wisdom suggests that companies with higher ESG activities increase their market valuation against their recorded net assets (i.e., Di Simone et al., 2022) even though the effect may exacerbate the misvaluation (Bofinger et al., 2022). Ferrell et al. (2016) also note that well-governed firms with fewer agency costs can take more full advantage of the positive valuation effects of CSR activities. Leverage, defined as the ratio of long-term debt to the book value of equity, aims at capturing financial constraints that may inhibit CSR activities. The impact of higher leverage on CSR appears to be negative, either by inhibiting directly sustainable investments (Hong et al., 2012) or by increasing the cost of debt (i.e., Goss & Roberts, 2011; Zerbib, 2019). Finally, to account for diversity in operations, firms are classified in a specific industry based on the Kenneth French 12-industry classification.

The PSM routine is implemented in two stages. First, we run a binary logit regression, where the "treatment variable" is violator status (i.e., a dummy equal to 1 for firms committing environmental violations and 0 otherwise) for each year separately. The first-stage regression produces the propensity score as the fitted probability of being a violator. Second, we use a k-nearest neighbor algorithm with $k = 1$ to match the firms based on the nearest propensity scores. To facilitate efficient matching, we demand that the difference in the logits of the propensity scores for pairs of individuals from the two groups be less than or equal to 0.5 times the pooled estimate of the common standard deviation of the logits of the propensity scores. The procedure is repeated for every year with annual replacement.

The ensuing sample comprises 1120 CSR reports published by 441 unique firms (see Table 1) over the period 2008–2022. This suggests, on average, 2.53 CSR reports per firm. While this number is sufficient for assessing *cross-sectional* differences between violator and nonviolator firms, in subsequent DiD analysis we require a longer time-series dimension. We, therefore, condition a subsample of 475 reports for 79 unique firms attaining, on average, six consecutive annual reports per firm, and a median number of reports before violation being 3.01. Only the companies that changed status from nonviolator to violator are included in the subsample.

4.3 | NLP variables

We apply standard NLP routines to quantify textual information for those firms/years in which we obtain CSR reports. Below, we describe these procedures.

4.3.1 | Text extraction

CSR reports are obtained as Portable Document Format (pdf) documents with various degrees of structure. To obtain useful information from the pdf, we transcribe the textual content to raw text and apply a tokenizer, splitting paragraphs and sentences into individual words or other nonwhitespace. This generates a "bag-of-words", a simplifying representation in which the linguistic information of text is represented by the set of its words without punctuation or order. Furthermore, parts of speech that are necessary for grammar but do not convey information, known as "stop words" are removed to reduce processing time. Examples of stop words include, but are not limited to, definite and indefinite articles (i.e., "the", "a" and "an"), adverbs (i.e., "because"), or prepositions (i.e., "until"). While stop words are completely eliminated, informative words are allowed multiplicity. The resulting set of words is

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TABLE 1 Corporate-social responsibility (CSR) reporting frequency and sample selection.

This table outlines the sample selection process. Panel A gives the details of the matching procedure. For each year, for the period 2008–2022, we estimate the equation propensity-score methodology (PSM): $logit(P(Viol = 1)) = a + b_1 Age + b_2 Size + b_3 ROE + b_4 Leverage + b_5 M/B + b_6 D/E + b_7 Intan + Sum(b_i Industry_i) + \varepsilon$ to predict the probability of treatment (violation) and match treated units to untreated units with *k*-nearest neighbor = 1 and caliber = 0.5. The process is repeated every year with annual replacement. Panel B provides a summary of matched pairs, number of unique firms and total number of reports for Violator versus Nonviolator firms. Panel C presents the annual breakdown of CSR report availability and provides statistics for differences in the mean, median and variance of the two groups. The ***, ** and * superscripts signify statistical significance at the 1%, 5% and 10% limits, respectively. All variables are defined in Appendix A.

Panel A: Propensity-score matching procedure			
Distance metric	Logit of PSM	Method	*k*-Nearest neighbors
Treatment variable	Environmental violation	Control ratio for *each year*	1 ($k = 1$)
Matched pairs with *Annual Replacement*	993	Caliber	0.5

Panel B: Reporting frequency all years |*N. of unique firms*: 441| Period: 2008–2022					
	Total (N.)	**Nonviolators (N.)**	**Violators (N.)**	**Difference (N.)**	**Violators/ nonviolators**
All year-matched observations	1986	993	993	0	
Available CSR reports	1120	482	638	156	
N. of unique firms	441	237	204		
N. of reports/firms	2.53	2.03	3.12	1.09	1.53

Panel C: Reporting frequency for matched sample per year					
		Available/collected CSR reports			
Year	**Matched pairs (N.)**	**Nonviolators (N.)**	**Violators (N.)**	**Difference (N.)**	**Violators/nonviolators (%)**
2008	60	10	15	5	50.00
2009	58	14	16	2	14.29
2010	60	16	37	21	131.25
2011	64	22	43	21	95.45
2012	68	31	40	9	29.03
2013	68	32	48	16	50.00
2014	76	36	55	19	52.78
2015	72	38	51	13	34.21

(Continues)

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TABLE 1 (Continued)

Panel C: Reporting frequency for matched sample per year					
		Available/collected CSR reports			
Year	Matched pairs (N.)	Nonviolators (N.)	Violators (N.)	Difference (N.)	Violators/nonviolators (%)
2016	68	37	54	17	45.95
2017	62	28	32	4	14.29
2018	64	43	38	−6	−0.11.6
2019	61	29	34	5	17.24
2020	73	55	58	3	5.45
2021	67	46	57	11	23.91
2022	72	45	60	15	33.33
Mean		32.13	43.53	10.33** (<0.04)	(t Statistics = −14.02)
Median		32.00	43.00	11.00* (0.07)	(Mann–Whitney statistics = 69.0)
SD		12.76	14.02	1.49 (0.418)	(Bartlett's U statistics = 0.66)

lemmatized, allowing the grouping of inflected forms of a word (i.e., "environmental" vs. "environmentally") to be analyzed as a single word. All words are converted to lowercase. Allowing a multiplicity of words and applying lemmatization creates a more meaningful representation of word frequency, an important input to the training classifiers.

4.3.2 | Environmental score

Environmental Score Analysis (ESA) aims to measure the environmental content of a CSR report in relation to other (nonidentified) themes in the text. While environmental content is expected to be a significant part of a sustainability report, the degree of emphasis given and space allocated to environmental issues remain discretionary and may reflect managerial decisions. ESA aims to gauge degrees of environmental reporting across firms and time, for violators or nonviolators, before and after a violation.

Our ESA analysis rests on the lexicon approach, in which a prearranged dictionary of environmental terms (the Lexicon) is used to score a CSR report based on the cumulative frequency of lexical entries found in the document. We obtain a Lexicon for environmental terms from DiCoEnviro (L'Homme, 2012), a multilingual technology resource that is publicly available.[5]

Unlike most terminological repositories DiCoEnviro is curated by linguists and is a priori designed for NLP. The English version of the Lexicon consists of 923 tokens. After lemmatizing and converting to lowercase, we count the number of times (if any) a lexical token appears in a

[5]DiCoEnviro can be accessed at the following link: https://olst.ling.umontreal.ca/dicoenviro/moteur/search-enviro.cgi?ui=en&mode=terme&lang=fr&prec=exact&equi=1&rq=

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CSR report. Then, we sum the counts and divide the total by the number of words in the text. The result gives a quantifiable measure of the environmental emphasis of a CSR report and is used thereon as our ES variable. ES values range from 0 to 1 and can be viewed as frequentist probabilities. The economic interpretation of ES is nevertheless tricky. For one, the plausible range of ES values is truncated: a value of 1 (or close to 1) is not attainable because such text would be illegible when reconstructed in sentences and paragraphs. An ES value of zero is theoretically possible, but unlike a report devoid of any environmental terms. Furthermore, looking at the central location of the distribution, an ES value of 0.5 does not signify environmental "neutrality": It means that half of the CSR report text consists of environmental terms, an enormous amount given that CSR reports are thematically diverse. Because we do not have a meaningful benchmark for evaluating ES values in absolute terms, we focus on assessing variation in ES values across firms and time, controlling for other factors.

4.3.3 | Readability score (RS)

Our second NLP variable is an RS assigned to each CSR report. An RS measures the degree of understanding of a text and roughly corresponds to the grade levels of the US education system required to comprehend the text. When applied to CSR reports, it gauges the degree to which the report is accessible to average readers. A CSR report with relatively low RS is understood to be more difficult to read and may include technical terms or longer sentences. Obfuscating the text of corporate reports allows management to pass false claims or conceal adverse facts. The extensive use of technical jargon, including environmental terms, may suggest "greenwashing". As such, the RS score is particularly important when jointly evaluated with our ES because it helps disentangle meaningful environmental content from greenwashing. Several tools are easily applied for obtaining an RS using the Textstat library in Python software. We rely on the Flesch Reading-Ease (FRE) test, which approximates readability as a function of total words in a sentence and total syllables in a word.[6] Due to the richness and convenient application of Textstat, we obtain alternative measures of readability (including the popular Coleman-Liau Index) and correlate those with FRE scores. We find FRE to be a robust and reliable measure of readability.

4.3.4 | Sentiment analysis

SA is a commonly applied ML tool that analyzes text for polarity, from positive to negative. In our case, the procedure output assigns a value between 0 and 1 to each CSR report, representing the probability of the report being positive. The economic interpretation is that the closer the score is to 1 the more positive the report will likely be. We note that CSR reports are corporate documents issued by firms to communicate their best efforts and their impact on the environment and community. While increasingly in demand, CSR reporting is optional. Firms that choose to issue a CSR report, do so to improve their outlook for external stakeholders, including providers of finance, customers, analysts and regulators. For this reason, the content

[6]Note that to apply the FRE test, the sentence structure of the CSR text must be maintained. Therefore, while the CSR report is reduced to text, it is not tokenized.

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of a CSR report is expected to be, and is, almost by definition, positive. It is nevertheless valuable to assess a cross-sectional variation in positive sentiment and to compare degrees of positiveness between violators and nonviolators.

Our SA relies on the corpus approach, in which an ML algorithm (the classifier) is trained to categorize novel text into positive or negative. The classifier "learns" to distinguish a positive text from a negative text by looking at many validated examples. The set of documents representing examples of positive or negative text is called the corpus.[7] After learning is completed to a high degree of accuracy, the trained classifier is applied to predict sentiment on each CSR report. We rely on the Amazon Unlocked Mobile data set to train the classifier, a widely used SA corpus available at Kaggle. The set comprises 400,000 reviews of mobile phones; each ranked in terms of positiveness on a scale of 1–5. Using the same techniques outlined for text extraction, we tokenize the set to remove punctuation and stop words, lemmatize, and convert all words to lowercase. We remove all instances with a rating of three (3) signifying neutral reports, neither positive nor negative, and we encode the remaining set by assigning a value of 1 (positive) to ratings above three or a value of zero (negative) otherwise. In the next step, we transform the textual information into a meaningful representation of numbers in which each word, for each review, is indexed to a vocabulary, and the frequency of each word concerning this index is calculated (count vectorization). The procedure is further refined as follows: First, we remove words appearing in fewer than five separate reviews. As the reviews are completed ad hoc by online customers, they contain words with misspellings, words augmented by numbers, misnomers, or other tokens that are not intelligible to NLP for purposes of lemmatization. Because such input is idiosyncratic to the person completing a single review, demanding that a minimum of five separate reviews include a specific token alleviates the possibility of featuring meaningless information. Eliminating redundant tokens also preserves degrees of freedom. Finally, to provide a weight on how important a word is to a review, we apply the method of term frequency-inverse document frequency (tf-idf), which compares the number a word appears in a review with the number of reviews the word appears in. As the tf-idf statistic decreases by how many times a word appears in the corpus, just like stop words, it reduces the bias of assigning greater weight to commonly used words. Tf-idf values are stored in a matrix in which each row represents a review, and each column represents a word. The matrix is sparse, that is, mostly populated by zeros because there are many more words than reviews. Since we already know whether a review is positive (assigned a value of 1 for scores higher than 3) or negative (assigned a value of 0), we now run a binary logistic regression model that outputs the estimated coefficients of each word (weighted by its tf-idf value) as explanatory variables of the sentiment score. The model's accuracy is tested out-of-sample by splitting the corpus into a training set (80% of the reviews) and a test set. Finally, we apply the estimated coefficients to the vectorized representation of each CSR report to predict the probability of the report being positive.

4.4 | Compustat variables

We match NLP variables with additional annual Compustat variables using ticker/CUSIP and year as identifiers. According to our proposed Hypotheses H5–H7, these variables capture

[7]In general, a corpus is any collection of authentic text, meaning text written by a native of the language. Different corpora support different ML objectives, beyond SA.

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board characteristics, ownership structure and international presence (see data description in Appendix A).

5 | EMPIRICAL RESULTS

5.1 | Descriptive statistics

Table 1, Panel A gives the details of the PSM procedure. The total number of matched pairs, extracted with annual replacement over the period 2008–2022 is 993. Panel B gives a breakdown of the total number of matched observations, collected CSR reports and the number of unique firms per violator status. We find that the public availability of CSR reports for violator firms is significantly higher than that of the nonviolator firms. For an equal number of (matched) firms throughout the same period, violator firms appear to volunteer significantly more CSR information, in terms of a number of publicly available CSR reports, by a factor of 1.53.

Panel C breaks down the number of CSR reports collected for each year of the sample for violator/nonviolator firms. Consistent with the findings of Panel B, violator firms appear to volunteer more CSR reports in (nearly) all years of the sample with the % of higher frequency ranging from 5.45% to 131.3% and an overall rate of 39.8%. The mean and median tests of differences in the reporting frequency of the violator versus nonviolator firms are statistically significant and the two samples are shown to have the same variance.

Throughout our search, we identify several ways in which companies choose to volunteer or conceal more CSR information. First, companies differ in their decision to issue a stand-alone CSR report, or not. Because the collection of CSR reports is performed retrospectively, we can only be reasonably certain that a CSR report has NOT been issued in the year of search or near past (last 2 years). Second, there is significant variation in the time-series public availability of CSR reports. While some firms maintain a public record of CSR reports stretching over several years, others maintain a more conservative outlook keeping online only the latest reports. Third, firms may choose to publicly advertise the issuance of a CSR report but only provide access to the report for their shareholders, or other insiders. Finally, some CSR reports are publicly released but they are password protected or encrypted against NLP/semantic analysis.

Given the above set of choices, our first finding suggests that violator firms may voluntarily disclose more CSR information in an effort to maintain a better public outlook in contrast to their environmental actions. This is in line with Patten (2002), who suggests that companies with poor environmental performance are likely to volunteer more information. At the same time, our findings contradict Mahoney et al. (2013), who argue that companies with better CSR performance are more likely to issue stand-alone CSR reports, and to publish a higher number of reports (Uyar et al., 2020). These studies, however, use composite ESG scores, which are more noisy and not independently verifiable.

Nevertheless, at this point, we do not know, what the content of the CSR report is and whether it relates to environmental disclosure at all. Neither can we comment on the tone or readability of the report as means of attention deflection, decoupling and greenwashing.

Table 2 gives the descriptive statistics of the variables included in the sample. Panel A shows the NLP variables extracted from the CSR reports with respect to environmental focus, readability and positiveness. The interactive terms between environmental focus times

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TABLE 2 Descriptive statistics.

This table gives descriptive statistics of the explanatory variables used in the analysis. Panel A gives descriptive statistics for natural-language processing variables. Panel B provides Board composition characteristics, including the total number of directors, the % of the board composition comprising women directors and the number of outside directors. Panel C provides a breakdown between firms based solely in the United States versus firms founded in the United States but maintaining significant international presence. For those companies having an international presence, the panel gives the % of income generated from offshore operations and the % of total tax paid in foreign jurisdictions. Panel D gives a breakdown of the Kenneth French 12-industry classification. Panel E gives the corresponding breakdown for ownership structure. All variables are defined in Appendix A.

Panel A: Natural-language processing variables

	N	Mean	SD	5pct	25pct	Median	75pct	95pct
N-pages	1120	63	51	8	28	51	82	162
Total N. of Words	1120	22,536	21,722	1764	7924	16,646	29,834	60,547
Readability Score	1120	50.04	26.34	24.72	36.10	47.40	56.40	82.50
Positiveness Score	1120	0.807	0.077	0.665	0.760	0.820	0.864	0.910
Environ. Score	1120	0.179	0.030	0.138	0.163	0.179	0.195	0.220
*Envir. * Read*	1120	8.997	5.463	4.138	6.350	8.121	9.945	16.079
*Envir. * Pos*	1120	0.144	0.028	0.105	0.128	0.143	0.161	0.183

Panel B: Board characteristics

	N	Mean	SD	5pct	25pct	Median	75pct	95pct
Total N. of Directors	1120	10.30	2.11	7.00	9.00	10.00	12.00	14.00
Women Directors (N.)	1120	2.12	1.37	0	1.00	2.00	3.00	5.00
(Pct.)		(20%)	(12%)	(0%)	(11%)	(20%)	(27%)	(42%)
Outside Directors (N.)	1120	8.88	2.12	5.00	7.00	9.00	10.00	12.00
(Pct.)		(86%)	(8%)	(71%)	(82%)	(89%)	(91%)	(92%)

Panel C: International representation

	Local	Global	Total
N. firm/year observations	739	381	1120
%	66	34	100
% Of Foreign Income to Total Income	0	53	100
% Of Foreign Tax to Total Tax	0	69	100

Panel D: Industry classification

	N.Firm/Year	(%)
Energy	208	19
Materials	203	18

Panel E: Ownership

	N	(%)
Principal Shareholder	395	35
Institutions Dominant	182	16

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TABLE 2 (Continued)

Panel D: Industry classification			Panel E: Ownership		
	N.Firm/Year	**(%)**		***N***	**(%)**
Industrials	167	15	*Indexed Stock*	276	25
Consumer Discr.	112	10	*Founder Firm*	73	7
Consumer Staples	69	6.1	*Family Firm*	54	5
Health Care	73	6.5	*Mixed Ownership*	38	3
Financials	50	4.4	*Widely Held*	57	5
IT	66	5.9	*Controlling Shareholder*	46	4
Communication	20	1.8			
Utilities	84	7.5			
Real Estate	37	3.3			
Other	32	2.9			
Total	1120	100		1120	100

readability and environmental focus times positiveness are also shown. As a measure of report size, the table provides the number of pages and words within the reports.

The interpretation of NLP variables is tricky, and to some extent, not very useful when looked at in isolation. For example, there is no reference point to assess whether a mean environmental score of 0.179 is high or not, particularly given the diverse scope of a CSR report. At the same time, a CSR report is, by definition, a positive and fairly sophisticated document and these characteristics are reflected in both the positiveness score and RS we obtain. Of more interest is the cross-sectional variation of these variables and mean differences between groups of violator/nonviolator firms.

Panel B provides statistics on board representation variables, in particular, the number of directors (size), the % of women directors (diversity) and the number of outside directors (independence). The average number of directors appears to be just above 10.3 extending up to 14. Female representation is at a mean of 20% suggesting, on average, one female director versus five male directors on a typical board of 10. The range of female representation extends from 0% to 42% suggesting in some cases an "even" board, but never a majority female board. The high number of outside directors (on average 8.8 on an average total of 10) suggests strong independence, even though special cases of CEO duality may negate this. Across the entire range of board size and number of outside directors, at least one person on the board remains an insider, which is not unexpected.

Panel C gives information regarding the international presence of firms comprising the sample. We employ two measurable proxies for international operations: the first being the % of foreign income to total income and the second being the % of foreign tax to total tax. In each variable, a measurement of zero suggests no international presence. If at least one of the two is nonzero the company is classified as an international firm, otherwise domestic. The breakdown suggests that 66% of our firm/year observations are domestic firms and 34% are international. Importantly, for those international firms, approximately 53% of their income comes from

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overseas operations while 69% of their tax is paid to foreign jurisdictions. We acknowledge that these levels of financials may as well be the result of tax optimization practices in which taxable income is transferred to lower tax rate destinations, without reflecting the true scope of international operations. To that end, we only use "international presence" as a categorical variable, without further quantifying the international presence in accordance with foreign income or tax.

Panel D gives the industry breakdown. Approximately 52% of the firms are classified as "Energy", "Materials" or "Industrials". Looking at our initial sample selection process, we remind that our starting point is coverage for environmental violations in the KLD/MSCI database. As expected, such coverage is more heavily focused on companies in *environmentally taxing* industries. On the same note, financials, information technology firms and communications make up less than 12% of the sample combined. Even so, the sample is fairly well represented by the 12 industries proposed by Kenneth French with no missing classification. We thereon include industry fixed effects in the multivariate analysis.

Finally, Panel E provides information regarding the ownership structure of the firms. While there are eight categories of ownership, for purposes of further analysis, we group firms identified as "Institutions Dominant" or "Indexed Stock" or "Mixed Ownership" or "Widely Held" as higher dispersion firms, and firms classified as "Founder Firm" or "Family Firm" or having a "Principal Shareholder" or a "Controlling Shareholder" as lower dispersion firms. The resulting dummy variable is well balanced: aggregating the percentages of the different categories suggests that 49% of the firms fall under the classification of high dispersion, while 51% of the firms fall under the classification of low dispersion, a nearly 50–50 split.

5.2 | Univariate analysis

Table 3 presents group differences between violator/nonviolator firms regarding CSR textual characteristics, board composition, international status and ownership dispersion. Panel A presents mean differences of continuous variables. For each group, for each variable of interest, we calculate the mean statistic and then subtract the mean of the group of nonviolator firms from the mean of the group of the violator firms. To test for statistical significance, we apply the Mann–Whitney test assuming group independence. Panel B presents the analysis for categorical variables. For each group of violator–nonviolator status, we count the instances of international (domestic only) presence and low (high) ownership dispersion. The statistical significance for differences in frequency is assessed with the χ^2 test.

We find that the violator firms populate their CSR reports with greater amounts of environmental content, and the difference is statistically significant at less than 1%. The finding is in line with the theoretical premises of greenwashing via decoupling (Walker & Wan, 2012) in which environmental offenders project a textual image of sustainability that is not in line with their environmental actions. Higher ESs for violator firms persist when these interact with positiveness and readability. The violators tend to publish longer reports (the difference is statistically significant at less than 1%) and more positive reports (the difference is significant at 1%). We find little difference, when tested on a stand-alone basis, between readability of CSR reports of violator and nonviolator firms. Looking at board composition effects, we show that violator firms tend to have smaller boards, less diversity and less independence, and the results

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TABLE 3 Univariate analysis.

This table presents results from univariate analysis. Panel A presents differences of dependent variables between violator and nonviolator firms. The statistical significance of the differences is evaluated using the Mann–Whitney test, assuming independence. Violators are found to present in their corporate-social responsibility reports higher Environmental Scores and the result is significant at less than 1%. Higher environmental scores for violator firms persist when those are combined (interacted) with positiveness scores and readability scores. Furthermore, violator firms have smaller boards, represented by fewer women and outside directors. In Panel B a similar analysis is performed for categorical groups. Violator firms appear to have weaker international outlook (significant at less than 1%) and more concentrated ownerships. The statistical significance of differences in categorical variables is assessed through the χ^2 test. The ***, ** and * superscripts signify statistical significance at the 1%, 5% and 10% limits, respectively. All variables are defined in Appendix A.

Panel A: Continuous variables

	N	Violator	Nonviolator	Difference	*p* Value
Environ. Score	1120	0.185	0.175	0.010***	(<0.001)
Readability Score	1120	47.400	45.950	1.450	(0.114)
Positiveness Score	1120	0.823	0.816	−0.007**	(0.012)
N-pages	1120	57	46	11***	(<0.001)
Total N. of Words	1120	20,304	15,091	5213***	(<0.001)
*Environ. * Pos*	1120	8.651	7.733	0.919**	(0.018)
*Envir. * Read*	1120	0.147	0.142	0.005**	(<0.015)
N. of Directors	1120	10.230	10.438	−0.207**	(0.012)
% of Women Dir.	1120	0.193	0.210	−0.017***	(<0.001)
N. of Outside Dir.	1120	8.797	9.035	−0.238*	(0.083)

Panel B: Categorical variables

	N	Violator	Nonviolator	Odds ratio	Fisher exact *p* value
International	381	179 (47%)	202 (53%)		
Domestic	739	459 (62%)	280 (38%)	1.84***	(<0.001)
Total	1120				
High Own. Dispersion	625	292 (47%)	333 (53%)		
Low Own. Dispersion	495	346 (70%)	149 (30%)	2.64***	(<0.001)
Total	1120				

are statistically significant at less than 1% for women directors, 5% for size of the board and 10% for outside directors. These findings largely confirm Hypothesis H5, which states that board characteristics affect the probability of a firm being an environmental violator. Finally, we confirm that violator firms have a more limited international outlook and have a more concentrated ownership, confirming H6 and H7 respectively.

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5.3 | Multivariate analysis

To test for the joint effect of explanatory variables, we run a multivariate logit regression for the following specification:

$$\begin{aligned} p(Viol_{i,t}) = \alpha_0 &+ \beta_1 Env\text{-}Sc_{i,t} + \beta_2 Read\text{-}Sc_{i,t} + \beta_3 Pos\text{-}Sc_{i,t} + \beta_4 Length_{i,t} \\ &+ \beta_5 Env{*}Pos_{i,t} + \beta_6 Env{*}Read_{i,t} + \beta_7 NDir_{i,t} + \beta_8 \%WomD_{i,t} \\ &+ \beta_9 NDout_{i,t} + \beta_{10} IntrD_{i,t} + \beta_{11} DispD_{i,t} + f_{ind} + f_{year} + \epsilon_{i,t}. \end{aligned} \quad (1)$$

The dependent variable is the likelihood of being a violator firm while all regressors are as described in Sections 4.3 and 4.4 and as defined in Appendix A. We also include industry and year fixed effects to control for confounding factors. Table 4 gives the correlation matrix in which the pairwise associations of the individual variables are measured, while Table 5 gives the regression results.

The output of the logistic regression gives the estimated coefficients, the average marginal effects and the corresponding odds ratios. For marginal effects and odds ratios, we also provide lower (2.5%) and upper (97.5%) bounds. Since the dependent variable is the log odds of a firm being an environmental violator the estimated coefficients represent the change of the log odds given the incremental increase in the given explanatory variable, all other things constant. The odds ratio equals the exponent of the estimated coefficient providing a more straightforward way of looking at how, ceteris paribus, the odds of the outcome (dependent) variable change for a one-unit increase in the corresponding predictor variable. An odds ratio of more (less) than one suggests a positive (negative) association between the dependent variable and the corresponding regressor. Finally, marginal effects directly give how the predicted probability of being an environmental violator increases (decreases) for a one-unit increase in any predictor variable, holding all other variables constant.

We find that violator firms are positively associated with higher ESs, more positive content and larger reports, and negatively associated with the interaction of environmental content and readability. These results suggest that violator firms may use the information content of their CSR reports to obfuscate (decouple) their true environmental performance. In doing so, they volunteer more information (longer reports) with stronger and more positive environmental content, which nevertheless, is less readable. This is consistent with Higgins et al. (2020), who argue that a lack of clarity and accuracy in what is being reported could indicate duplicity and hypocrisy.

Furthermore, violator firms are found to be associated with smaller boards, limited representation of female directors, fewer outside directors and no international presence. It is interesting to compare the results of the multivariate regression with the univariate tests of association, as the joint assessment of individual predictors largely corrects for common trends and latent effects. While the interaction between ES and readability appears initially higher for violator firms, when the variable is tested simultaneously with other predictors, it reverts to the predicted negative sign.

5.4 | Change in violator status and modified reporting

Both univariate and multivariate tests reveal that violator firms are more likely to exploit CSR reporting to project an image of ecofriendliness and sustainability, even as they commit

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TABLE 4 Pearson correlation matrix.

This table gives the Pearson correlation matrix. Figures in brackets are *p* values of statistical significance. All variables are defined in Appendix A.

		(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
(1)	*Env-Sc*	1	0.07	−0.01	−0.02	−0.03	0.34	0.87	−0.12	−0.08	−0.12	−0.02	0.00
			(0.01)	(0.77)	(0.42)	(0.28)	(0.00)	(0.00)	(0.00)	(0.01)	(0.00)	(0.41)	(0.88)
(2)	*Read-Sc*		1	−0.02	0.00	0.01	0.95	0.05	0.00	−0.05	0.00	−0.06	−0.02
				(0.46)	(0.89)	(0.63)	(0.00)	(0.07)	(0.89)	(0.09)	(0.89)	(0.03)	(0.57)
(3)	*Pos-Sc*			1	−0.08	−0.12	−0.02	0.48	0.00	−0.04	0.00	−0.02	−0.01
					(0.00)	(0.00)	(0.53)	(0.00)	(0.92)	(0.19)	(0.95)	(0.44)	(0.68)
(4)	*N-pages*				1	0.91	−0.01	−0.06	0.07	0.13	0.05	0.07	0.02
						(0.00)	(0.65)	(0.03)	(0.03)	(0.00)	(0.08)	(0.02)	(0.43)
(5)	*N-words*					1	−0.01	−0.08	0.10	0.11	0.09	0.10	0.01
							(0.77)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.67)
(6)	*Env × Pos*						1	0.29	−0.04	−0.07	−0.04	−0.07	0.00
								(0.00)	(0.22)	(0.02)	(0.25)	(0.02)	(0.92)
(7)	*Env × Read*							1	−0.11	−0.09	−0.10	−0.03	0.00
									(0.00)	(0.00)	(0.00)	(0.32)	(0.99)
(8)	*NDir*								1	0.27	0.93	0.12	0.10
										(0.00)	(0.00)	(0.00)	(0.00)
(9)	*%WomD*									1	0.23	0.12	0.12
											(0.00)	(0.00)	(0.00)

(Continues)

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TABLE 4 (Continued)

		(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
(10)	*NDOut*										1	0.12	0.13
												(0.00)	(0.00)
(11)	*IntrD*											1	0.09
													(0.00)
(12)	*DispD*												1

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TABLE 5 Multivariate analysis.

This table gives the results from multivariate analysis. We estimate the equation: $p(Viol_{i,t}) = \alpha_0 + \beta_1 Env\text{-}Sc_{i,t} + \beta_2 Read\text{-}Sc_{i,t} + \beta_3 Pos\text{-}Sc_{i,t} + \beta_4 Length_{i,t} + \beta_5 Env * Pos_{i,t} + \beta_{6,t} Env * Read_{i,t} + \beta_7 NDir_{i,t} + \beta_8 \% WomD_{i,t} + \beta_9 NDout_{i,t} + \beta_{10} IntrD_{i,t} + \beta_{11} DispD_{i,t} + f_{ind} + f_{year} + \epsilon_{i,t}$, where the dependent variable is the log odds of being a violator with probability p versus not being a violator with probability $1 - p$. All regressors are defined in Appendix A. The table gives the estimated coefficients of the logistic regression, the average marginal effects (AMEs) and their lower/upper bounds and the average odds ratios (ORs) with corresponding lower (2.5%)/upper (97.5%) bounds. The estimated coefficients suggest the level of change in the log odds of being a violator for a unit of change on the corresponding explanatory variable. The AMEs are the mean of each individual observation's marginal effect directly measuring the change in probability of being a violator given an instantaneous change in a specific explanatory variable, with all other variables assumed constant. The ORs are obtained by exponentiating the estimated coefficients and can be interpreted as the change in the OR for an increase in one unit of the explanatory variable *relative* to no change. The ***, ** and * superscripts signify statistical significance at the 1%, 5% and 10% limits, respectively.

		Marginal effects			Odds ratios		
	Coefficients	**AME**	**Lower**	**Upper**	**OR**	**2.5%**	**97.5%**
Env-Sc	0.555***	0.11	−0.06	0.27	1.74	1.09	2.77
	(<0.01)						
Read-Sc	0.027	0.01	0.00	0.02	1.027	1.026	1.028
	(0.15)						
Pos-Sc	0.071**	0.01	−0.02	0.05	1.07	0.95	1.21
	(0.034)						
N-words	0.085***	0.02	0.00	0.04	1.09	1.05	1.13
	(<0.01)						
Env × Pos	0.153	0.03	−0.05	0.01	1.17	0.85	1.78
	(0.192)						
Env × Read	−0.404***	−0.08	−0.25	0.09	0.67	0.37	1.22
	(<0.01)						
NDir	−0.106***	−0.02	−0.03	−0.01	0.90	0.89	1.04
	(<0.01)						
%WomD	−0.255***	−0.05	−0.08	−0.02	0.77	0.55	1.09
	(<0.01)						
NDOut	−0.231***	0.04	0.01	0.08	1.26	0.09	1.38
	(<0.01)						
IntrD	−0.333***	−0.06	−0.12	−0.01	0.72	0.68	0.76
	(<0.01)						
DispD	−0.001	0.00	−0.03	0.02	0.99	0.88	1.12
	(0.876)						

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TABLE 5 (Continued)

		Marginal effects			Odds ratios		
	Coefficients	AME	Lower	Upper	OR	2.5%	97.5%
Industry fixed effects (FE)	YES	YES	YES	YES	YES	YES	YES
Year FE	YES	YES	YES	YES	YES	YES	YES
Maximum likelihood pseudo-R^2	14.45%						
N. of observations	1120						

punishable environmental offenses. The finding is largely cross-sectional which leaves room for latent factors influencing the relationship between violator status and CSR reporting, if violator firms are different from nonviolator firms in ways we do not control. To strengthen our analysis, we thereon investigate within-firm differences in CSR reporting, subject to change in violator status. The analysis answers the question of whether firms change their CSR reporting after becoming violators and provides a clearer cause-and-effect relationship between change in status and CSR reporting. The within-firm analysis helps control for unobserved heterogeneity by comparing the same firms over time and further reduces the selection bias from factors that do not vary over time within each firm.

To test for modified reporting, we seek to construct a subsample of firms with an adequate number of nonmissing consecutive CSR reports split evenly before and after the violation. After consolidating the KLD/MSCI and Refinitiv databases, we obtain 475 observations corresponding to 79 unique firms satisfying all data requirements. The mean number of reports per firm is 6 and the median number of reports before (after) the violation is 3. Table 6, Panel A provides the reports/firms breakdown.

5.5 | Cosine similarity

In Panel B we initially examine whether the content of a firm's CSR report is significantly altered straight after the firm's transition to the violator status. To test for the overall degree of modification, we use the cosine similarity measure, a standard NLP technique that directly compares the similarity of two documents. In cosine similarity, each document is transformed into a multidimensional vector, where each dimension represents a word in the document and the value in that dimension represents the frequency of that word in the document. In NLP, the angle between these vectors gives the degree of similarity with 1 indicating identical documents and 0 suggesting nonsimilarity (orthogonality).

To detect significant report modification, we observe whether the persistence in cosine similarity between subsequent reports of the same firm declines after change in violator status. More specifically, for all firms, we compare the median of the year-to-year cosine similarity in the period in which a firm retains its status, with the median of the cosine similarity between the report issued in the year following the change in status. We find that companies significantly modify their CSR report after the year of changing status. While the median cosine similarity for all instances of retaining violator status is approximately 12%, the cosine

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TABLE 6 Change in violator status and modified reporting.

This table presents a within-firm analysis for differences in the content of corporate-social responsibility (CSR) reporting upon change of status from nonviolator to violator. Panel A describes the sample partition of firms with available CSR reports for consecutive years and the change in status transitioning from nonviolator to violator. The average number of reports per firm in the sample is 6.01 with an even split of three reports before transition and three reports after the transition. Panel B provides results regarding tests of difference in medians regarding the overall (cosine) similarity of CSR reports before violation and the CSR report following the violation. In particular, for all firms, we compare the median of the year-to-year cosine similarity in the period in which a firm retains its status, with the median of the cosine similarity between the report issued in the year following the change in status. The cosine similarity measure directly compares the similarity of two documents. Each document is transformed into a multidimensional vector where each dimension represents a word in the document and the value in that dimension represents the frequency of that word in the document. The angle between these vectors gives the degree of similarity with 1 indicating identical documents and 0 suggesting nonsimilarity (orthogonality). $\Delta S(_t)$ signifies the year after change of status to violator, while $RS(_{t-n})$ denotes *n prior years* of retaining status of nonviolator. The results in Panel B indicate that firms modify their CSR reports more significantly upon to becoming a violator. The ***, ** and * superscripts signify statistical significance at the 1%, 5% and 10% limits, respectively.

Panel A: Sample partition for difference-in-difference analysis

	N. of reports	***Unique firms***	***N. of reports/firms***
Initial sample	1120	441	2.53
Change of status from nonviolator to violator	475	79	6.01
Median N. of reports before violation			3.01

Panel B: Within-firm differences in CSR content under alternate violator status

	$\Delta S(_t)$ median	**$RS(_{t-n})$ median**	**Difference**	**Mann–Whitney *p* value**
Cosine similarity ($CSR_{i,t}$, $CSR_{i,t-1}$)	0.091	0.120	−0.029***	[<0.01]

similarity following the change in status drops to 9.1% and the difference is statistically significant at 1%.

5.6 | DiD analysis

Having established that CSR reporting changes right after a violation, we seek to determine the direction of change. Since we already know that ex post violator firms tend to report more aggressively in terms of environmental content, positiveness and length, it is interesting to examine whether the modification is enacted after (or caused by) transitioning to violator status. For this test, we implement a two-way fixed effect staggered DiDs methodology in which the ESs, and other NLP measures, are dependent variables, and the change to violator status represents the treatment. The staggered analysis is particularly suitable for this sample because

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the treatment (violation) happens at different years over firms.[8] In Table 7, Panel A, we estimate the following specification:

$$y_{i,t} = \alpha_{ind} + \lambda_t + \beta^{DD} D_{i,t} + \varepsilon_{i,t}, \quad (2)$$

where α_{ind} and λ_t are industry/year fixed effects and $D_{i,t}$ an interaction variable between the treated unit (D_i) and an indicator variable for observations following ($POST_t$) the treatment (violation). The dependent variable $y_{i,t}$ is substituted with the environmental score (*Env-Sc*), and in turn, readability (*Read-Sc*), positiveness (*Pos-Sc*), length (*N-words*), the interaction between environmental score and positiveness (*Env-Pos*) and the interaction between environmental score and readability (*Env-Read*).

The analysis confirms that companies modify their CSR reports following an environmental violation to significantly increase the amount of environmental content, positiveness and length. We are unable to confirm a within-firm negative relationship between change in violator status and readability and its interaction with ES.

We furthermore augment the DiD analysis by adding firm controls for additional robustness. While all firms included in the DiD sample are initially matched, extending their time series to include a sufficient number of reports may fall outside the matched period. Hence, we additionally estimate:

$$y_{i,t} = \alpha_{ind} + \lambda_t + \gamma^i \bar{X}_{i,t} + \beta^{DD} D_{i,t} + \varepsilon_{i,t}, \quad (3)$$

where $\bar{X}_{i,t}$ is a vector of firm characteristics, including Size, Profitability and Leverage (as defined in Appendix A), and all other variables as in Equation (1). Panel B shows that the DiD results survive the inclusion of additional firm controls. Overall, this section reinforces the idea that differences in CSR reporting between violator and nonviolator firms are initiated following an environmental transgression to counterbalance, or preempt negative publicity.

6 | CONCLUSIONS

We revisit the issue of alleged greenwashing, the practice by which firms' benevolent environmental claims are in contrast to their actual environmental performance. Although there are various channels through which these claims are communicated, we focus on CSR reports, as the medium for corporate disclosure displaying a comparatively higher level of accountability. At the same time, we note that CSR reporting remains largely unregulated and unaudited, providing corporations with significant flexibility in regard to the level of information they choose to divulge—both in terms of reporting frequency and content. We compare the CSR disclosure for firms with documented environmental violations versus those firms with a clean environmental record. Using standard NLP techniques, we provide quantitative measures of size, readability, environmental focus and positiveness of CSR disclosure.

We find that violator firms tend to issue longer, more positive and more frequently publicly available reports, and use these reports to broadcast environmental content that is more

[8]We thank an anonymous referee for suggesting this line of research.

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TABLE 7 Difference-in-difference (DiD) analysis.

This table gives the results of a staggered DiD analysis. In Panel A, we estimate $y_{i,t} = \alpha_{ind} + \lambda_t + \beta^{DD} D_{i,t} + \varepsilon_{i,t}$, where α_{ind} and λ_t are industry/year fixed effects (FE) and $D_{i,t}$ an interaction variable between the treated unit (D_i) and an indicator variable for observations following ($POST_t$) the treatment (violation). The dependent variable $y_{i,t}$ takes the value of environmental score (*Env-Sc*) (*column* 1) and in turn, readability (*Read-Sc*), positiveness (*Pos-Sc*), length (*N-words*), the interaction between environmental score and positiveness (*Env-Pos*) and the interaction between environmental score and readability (*Env-Read*). Panel B augments the analysis with firm-specific controls. We estimate $y_{i,t} = \alpha_{ind} + \lambda_t + \gamma^i \bar{X}_{i,t} + \beta^{DD} D_{i,t} + \varepsilon_{i,t}$ where $\bar{X}_{i,t}$ is a vector of firm characteristics, including Size, Profitability and Leverage. All variables are defined in Appendix A. The ***, ** and * superscripts signify statistical significance at the 1%, 5% and 10% limits, respectively.

Panel A: Staggered DiD with year and industry/fixed effects |*N. of observations* = 475|						
	***Env-Sc* (1)**	***Read-Sc* (2)**	***Pos-Sc* (3)**	***N-words* (4)**	***Env * Pos* (5)**	***Env * Read* (6)**
β^{DD}	0.175*** (<0.001)					
β^{DD}		0.449 (0.750)				
β^{DD}			0.804*** (<0.001)			
β^{DD}				0.917*** (<0.001)		
β^{DD}					3.328*** (<0.001)	
β^{DD}						0.142*** (<0.001)
Year FE	YES	YES	YES	YES	YES	YES
Industry FE	YES	YES	YES	YES	YES	YES
Adjusted R^2	0.23	0.22	0.28	0.31	0.79	0.22
N. of firms	79					
Reports per firm	6.01					

Panel B: Staggered DiD with year/industry FE and firm controls |*N. of observations* = 475|						
	***Env-Sc* (7)**	***Read-Sc* (8)**	***Pos-Sc* (9)**	***N-words* (10)**	***Env * Pos* (11)**	***Env * Read* (12)**
β^{DD}	0.172*** (<0.001)					
β^{DD}		0.449 (0.750)				
β^{DD}			0.768*** (<0.001)			

(Continues)

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TABLE 7 (Continued)

Panel B: Staggered DiD with year/industry FE and firm controls |*N. of observations* = 475|						
	***Env-Sc* (7)**	***Read-Sc* (8)**	***Pos-Sc* (9)**	***N-words* (10)**	***Env * Pos* (11)**	***Env * Read* (12)**
β^{DD}				0.997*** (<0.001)		
β^{DD}					3.304*** (<0.001)	
β^{DD}						0.092*** (<0.001)
Firm controls	YES	YES	YES	YES	YES	YES
Year FE	YES	YES	YES	YES	YES	YES
Industry FE	YES	YES	YES	YES	YES	YES
Adjusted R^2	0.27	0.22	0.34	0.35	0.78	0.18
N. of firms	79					
Reports per firm	6.01					

copious, but less readable. Issuing longer reports, more frequently, suggests that violator firms are willing to divulge more information in line with selective disclosure theory. Populating the report with greater amounts of environmental content, albeit with lower degree of readability, likewise, points out to attention deflection and decoupling. Furthermore, we show that reporting changes right after committing a violation. The cosine similarity test and differences-in-differences regressions highlight the within-firm differences in CSR reporting and provide a clearer cause-and-effect relationship between the change in status and CSR reporting controlling for all firm-specific effects.

Our study makes important contributions. First, we identify greenwashing in CSR reports using a novel methodology—NLP techniques, which allow analysis of the large body of textual data (Aziz et al., 2022). To the best of our knowledge, such analysis has not been done before to detect greenwashing. The use of NLP techniques offers significant advantages both in terms of the quantity of processed textual data as well as in the construction of original variables of interest extracted from text, such as environmental content, tone and readability. Second, in line with the literature that defines greenwashing as a gap between *verba et acta* (words and deeds), we compare what companies say with what companies do. Unlike previous studies, for example, Ioannou et al. (2022), that use direct measures of greenwashing—the difference between the constructs of policy and implementation, or the deviation between the reported and actual emissions (Kim & Lyon, 2015), we observe greenwashing "in action": the violator firms report more aggressively, more positively and volunteer more information to relay environmental content that is more copious but less readable. In this manner, while we maintain the conceptual framework of greenwashing, we contribute a novel way to detecting greenwashing in CSR reporting. Third, we catch the violators "red-handed": our study demonstrates that the CSR reporting changes straight after committing a violation.

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Overall, we find evidence that culpable firms may exploit the lack of regulation and employ CSR reports as a means of greenwashing. This raises the question of whether a statutory framework for CSR reporting is now needed. Despite arguments against mandatory disclosure, including significant adjustment costs and/or loss of competitive advantage, we take the view that the benefits will eventually outweigh the additional regulatory burden.

DATA AVAILABILITY STATEMENT

Financial data are obtained from financial databases described in detail in the text. The lexicon of environmental terms can be found at http://olst.ling.umontreal.ca/dicoenviro/dicoenviro-bilingue-fr.html. The corporate-social responsibility/sustainability reports used in NLP are obtained from publicly available corporate websites and stored locally due to the large size (in excess of 6 GB). They are available upon request. All data used in the study are publicly available from sources cited in the text. The list of companies in the sample is available upon request.

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REFERENCES

Abeysekera, A. P., & Fernando, C. S. (2020). Corporate social responsibility versus corporate shareholder responsibility: A family firm perspective. *Journal of Corporate Finance*, *61*, 101370. https://doi.org/10.1016/j.jcorpfin.2018.05.003

Albitar, K., Abdoush, T., & Hussainey, K. (2023). Do corporate governance mechanisms and ESG disclosure drive CSR narrative tones? *International Journal of Finance & Economics*, *28*(4), 3876–3890. https://doi.org/10.1002/ijfe.2625

Aziz, S., Dowling, M., Hammami, H., & Piepenbrink, A. (2022). Machine learning in finance: A topic modeling approach. *European Financial Management*, *28*(3), 744–770. https://doi.org/10.1111/eufm.12326

Baker, M., Egan, M. L., & Sarkar, S. K. (2022). *How do investors value ESG*? [NBER Working Paper 30708]. https://www.nber.org/system/files/working_papers/w30708/w30708.pdf

Bénabou, R., & Tirole, J. (2010). Individual and corporate social responsibility. *Economica*, *77*(305), 1–19. https://doi.org/10.1111/j.1468-0335.2009.00843.x

Berg, F., Kölbel, J. F., & Rigobon, R. (2022). Aggregate confusion: The divergence of ESG ratings. *Review of Finance*, *26*(6), 1315–1344. https://doi.org/10.1093/rof/rfac033

Bofinger, Y., Heyden, K. J., & Rock, B. (2022). Corporate social responsibility and market efficiency: Evidence from ESG and misvaluation measures. *Journal of Banking & Finance*, *134*, 106322. https://doi.org/10.1016/j.jbankfin.2021.106322

Boone, A. L., Casares Field, L., Karpoff, J. M., & Raheja, C. G. (2007). The determinants of corporate board size and composition: An empirical analysis. *Journal of Financial Economics*, *85*(1), 66–101. https://doi.org/10.1016/j.jfineco.2006.05.004

Borghesi, R., Houston, J. F., & Naranjo, A. (2014). Corporate socially responsible investments: CEO altruism, reputation, and shareholder interests. *Journal of Corporate Finance*, *26*, 164–181. https://doi.org/10.1016/j.jcorpfin.2014.03.008

Cai, Y., Pan, C. H., & Statman, M. (2016). Why do countries matter so much in corporate social performance? *Journal of Corporate Finance*, *41*, 591–609. https://doi.org/10.1016/j.jcorpfin.2016.09.004

Cespa, G., & Cestone, G. (2007). Corporate social responsibility and managerial entrenchment. *Journal of Economics & Management Strategy*, *16*(3), 741–771. https://doi.org/10.1111/j.1530-9134.2007.00156.x

Chae, J. (2005). Trading volume, information asymmetry, and timing information. *The Journal of Finance*, *60*(1), 413–442. https://www.jstor.org/stable/3694843

1468036x, 2025, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/eufm.12569 by CochraneChina, Wiley Online Library on [01/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Chatterji, A. K., Durand, R., Levine, D. I., & Touboul, S. (2016). Do ratings of firms converge? Implications for managers, investors and strategy researchers. *Strategic Management Journal*, *37*(8), 1597–1614. https://doi.org/10.1002/smj.2407

Chava, S. (2014). Environmental externalities and cost of capital. *Management Science*, *60*(9), 2223–2247. https://doi.org/10.1287/mnsc.2013.1863

Christensen, H. B., Hail, L., & Leuz, C. (2021). Mandatory CSR and sustainability reporting: Economic analysis and literature review. *Review of Accounting Studies*, *26*(3), 1176–1248. https://doi.org/10.1007/s11142-021-09609-5

Clarkson, P. M., Ponn, J., Richardson, G. D., Rudzicz, F., Tsang, A., & Wang, J. (2020). A textual analysis of US corporate social responsibility reports. *Abacus*, *56*(1), 3–34. https://doi.org/10.1111/abac.12182

Cornett, M. M., Erhemjamts, O., & Tehranian, H. (2016). Greed or good deeds: An examination of the relation between corporate social responsibility and the financial performance of US commercial banks around the financial crisis. *Journal of Banking & Finance*, *70*, 137–159. https://doi.org/10.1016/j.jbankfin.2016.04.024

Cronqvist, H., & Yu, F. (2017). Shaped by their daughters: Executives, female socialization, and corporate social responsibility. *Journal of Financial Economics*, *126*(3), 543–562. https://doi.org/10.1016/j.jfineco.2017.09.003

D'Amato, A., & Falivena, C. (2020). Corporate social responsibility and firm value: Do firm size and age matter? Empirical evidence from European listed companies. *Corporate Social Responsibility and Environmental Management*, *27*(2), 909–924. https://doi.org/10.1002/csr.1855

De Freitas Netto, S. V., Sobral, M. F. F., Ribeiro, A. R. B., & Soares, G. R. L. (2020). Concepts and forms of greenwashing: A systematic review. *Environmental Sciences Europe*, *32*(1), 19. https://doi.org/10.1186/s12302-020-0300-3

Dehejia, R. H., & Wahba, S. (2002). Propensity score-matching methods for nonexperimental causal studies. *Review of Economics and Statistics*, *84*(1), 151–161. https://doi.org/10.1162/003465302317331982

Di Simone, L., Petracci, B., & Piva, M. (2022). Economic sustainability, innovation, and the ESG factors: An empirical investigation. *Sustainability*, *14*(4), 2270. https://doi.org/10.3390/su14042270

Drobetz, W., El Ghoul, S., Fu, Z., & Guedhami, O. (2024). Institutional investors and corporate environmental costs: The roles of investment horizon and investor origin. *European Financial Management*, *30*(2), 727–769. https://doi.org/10.1111/eufm.12444

Edmans, A. (2011). Does the stock market fully value intangibles? Employee satisfaction and equity prices. *Journal of Financial Economics*, *101*(3), 621–640. https://doi.org/10.1016/j.jfineco.2011.03.021

Edmans, A. (2023). The end of ESG. *Financial Management*, *52*(1), 3–17. https://doi.org/10.1111/fima.12413

Ferrell, A., Liang, H., & Renneboog, L. (2016). Socially responsible firms. *Journal of Financial Economics*, *122*(3), 585–606. https://doi.org/10.1016/j.jfineco.2015.12.003

Fiechter, P., Hitz, J.-M., & Lehmann, N. (2022). Real effects of a widespread CSR reporting mandate: Evidence from the European Union's CSR Directive. *Journal of Accounting Research*, *60*, 1499–1549. https://doi.org/10.1111/1475-679X.12424

Font, X., Walmsley, A., Cogotti, S., McCombes, L., & Häusler, N. (2012). Corporate social responsibility: The disclosure–performance gap. *Tourism Management*, *33*(6), 1544–1553. https://doi.org/10.1016/j.tourman.2012.02.012

El Ghoul, S., Guedhami, O., Wang, H., & Kwok, C. C. Y. (2016). Family control and corporate social responsibility. *Journal of Banking & Finance*, *73*, 131–146. https://doi.org/10.1016/j.jbankfin.2016.08.008

Gillan, S. L., Koch, A., & Starks, L. T. (2021). Firms and social responsibility: A review of ESG and CSR research in corporate finance. *Journal of Corporate Finance*, *66*, 101889. https://doi.org/10.1016/j.jcorpfin.2021.101889

Goss, A., & Roberts, G. S. (2011). The impact of corporate social responsibility on the cost of bank loans. *Journal of Banking & Finance*, *35*(7), 1794–1810. https://doi.org/10.1016/j.jbankfin.2010.12.002

Greening, D. W., & Turban, D. B. (2000). Corporate social performance as a competitive advantage in attracting a quality workforce. *Business & Society*, *39*(3), 254–280. https://doi.org/10.1177/000765030003900302

Guo, R., Zhang, W., Wang, T., Li, C. B., & Tao, L. (2018). Timely or considered? Brand trust repair strategies and mechanism after greenwashing in China—From a legitimacy perspective. *Industrial Marketing Management*, *72*, 127–137. https://doi.org/10.1016/j.indmarman.2018.04.001

1468036x, 2025, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/eufm.12569 by CochraneChina, Wiley Online Library on [01/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Haji, A. A., Coram, P., & Troshani, I. (2023). Consequences of CSR reporting regulations worldwide: A review and research agenda. *Accounting, Auditing & Accountability Journal*, *36*(1), 177–208. https://doi.org/10.1108/AAAJ-05-2020-4571

Higgins, C., Tang, S., & Stubbs, W. (2020). On managing hypocrisy: The transparency of sustainability reports. *Journal of Business Research*, *114*, 395–407. https://doi.org/10.1016/j.jbusres.2019.08.041

Hinnen, G., Hille, S. L., & Wittmer, A. (2017). Willingness to pay for green products in air travel: Ready for take-off? *Business Strategy and the Environment*, *26*(2), 197–208. https://doi.org/10.1002/bse.1909

Hong, H., Kubik, J. D., & Scheinkman, J. A. (2012). *Financial constraints on corporate goodness* [Working Paper No. w18476]. National Bureau of Economic Research. http://www.nber.org/papers/w18476

Hong, H., & Shore, E. (2023). Corporate social responsibility. *Annual Review of Financial Economics*, *15*, 327–350.

Iliev, P., & Roth, L. (2023). Director expertise and corporate sustainability. *Review of Finance*, *27*(6), 2085–2123. https://doi.org/10.1093/rof/rfad012

Ioannou, I., Kassinis, G., & Papagiannakis, G. (2022). The impact of perceived greenwashing on customer satisfaction and the contingent role of capability reputation. *Journal of Business Ethics*, *185*, 333–347. https://doi.org/10.1007/s10551-022-05151-9

Jun, W., Shiyong, Z., & Yi, T. (2022). Does ESG disclosure help improve intangible capital? Evidence from A-share listed companies. *Frontiers in Environmental Science*, *10*, 858548. https://doi.org/10.3389/fenvs.2022.858548

Kim, E. H., & Lyon, T. P. (2015). Greenwash vs. brownwash: Exaggeration and undue modesty in corporate sustainability disclosure. *Organization Science*, *26*(3), 705–723. https://doi.org/10.1287/orsc.2014.0949

Knyazeva, A., Knyazeva, D., & Masulis, R. W. (2013). The supply of corporate directors and board independence. *Review of Financial Studies*, *26*(6), 1561–1605. https://doi.org/10.1093/rfs/hht020

L'Homme, M. C. (2012). Adding syntactico-semantic information to specialized dictionaries: An application of the FrameNet methodology. *Lexicographica*, *28*(2012), 233–252. https://doi.org/10.1515/lexi.2012-0012

Lang, M., & Stice-Lawrence, L. (2015). Textual analysis and international financial reporting: Large sample evidence. *Journal of Accounting and Economics*, *60*(2–3), 110–135. https://doi.org/10.1016/j.jacceco.2015.09.002

Larcker, D. F., Tayan, B., & Watts, E. M. (2022). Seven myths of ESG. *European Financial Management*, *28*(4), 869–882. https://doi.org/10.1111/eufm.12378

Lyon, T. P., & Maxwell, J. W. (2011). Greenwash: Corporate environmental disclosure under threat of audit. *Journal of Economics & Management Strategy*, *20*(1), 3–41. https://doi.org/10.1111/j.1530-9134.2010.00282.x

Mahoney, L. S., Thorne, L., Cecil, L., & LaGore, W. (2013). A research note on standalone corporate social responsibility reports: Signaling or greenwashing? *Critical Perspectives on Accounting*, *24*(4–5), 350–359. https://doi.org/10.1016/j.cpa.2012.09.008

Martin, W. C., Ponder, N., & Lueg, J. E. (2009). Price fairness perceptions and customer loyalty in a retail context. *Journal of Business Research*, *62*(6), 588–593. https://doi.org/10.1016/j.jbusres.2008.05.017

McGuinness, P. B., Vieito, J. P., & Wang, M. (2017). The role of board gender and foreign ownership in the CSR performance of Chinese listed firms. *Journal of Corporate Finance*, *42*, 75–99. https://doi.org/10.1016/j.jcorpfin.2016.11.001

Menla Ali, F., Wu, Y., & Zhang, X. (2023). ESG disclosure, CEO power and incentives and corporate risk-taking. *European Financial Management*, *30*(2), 961–1011. https://doi.org/10.1111/eufm.12447

Michelon, G., Pilonato, S., & Ricceri, F. (2015). CSR reporting practices and the quality of disclosure: An empirical analysis. *Critical Perspectives on Accounting*, *33*, 59–78. https://doi.org/10.1016/j.cpa.2014.10.003

Nagy, Z., Kassam, A., & Lee, L. E. (2016). Can ESG add alpha? An analysis of ESG tilt and momentum strategies. *The Journal of Investing*, *25*(2), 113–124.

Namkung, Y., & Jang, S. (2017). Are consumers willing to pay more for green practices at restaurants? *Journal of Hospitality & Tourism Research*, *41*(3), 329–356. https://doi.org/10.1177/1096348014525632

Nofsinger, J. R., Sulaeman, J., & Varma, A. (2019). Institutional investors and corporate social responsibility. *Journal of Corporate Finance*, *58*, 700–725. https://doi.org/10.1016/j.jcorpfin.2019.07.012

Orlitzky, M., & Benjamin, J. D. (2001). Corporate social performance and firm risk: A meta-analytic review. *Business & Society*, *40*(4), 369–396. https://doi.org/10.1177/00076503010400040

1468036x, 2025, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/eufm.12509 by CochraneChina, Wiley Online Library on [01/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Parguel, B., Benoit-Moreau, F., & Russell, C. A. (2015). Can evoking nature in advertising mislead consumers? the power of 'executional greenwashing'. *International Journal of Advertising*, *34*(1), 107–134. https://doi.org/10.1080/02650487.2014.996116

Pástor, Ľ., Stambaugh, R. F., & Taylor, L. A. (2022). Dissecting green returns. *Journal of Financial Economics*, *146*(2), 403–424. https://doi.org/10.1016/j.jfineco.2022.07.007

Patten, D. M. (2002). The relation between environmental performance and environmental disclosure: A research note. *Accounting, Organizations and Society*, *27*(8), 763–773. https://doi.org/10.1016/S0361-3682(02)00028-4

Principles of Responsible Investment. (2020). https://www.unpri.org/

SEC. (2020). *Recommendation from the investor-as-owner subcommittee of the SEC investor advisory committee related to ESG disclosure*. Accessed July 12, 2024 Available at: https://www.sec.gov/spotlight/investor-advisory-committee-2012/recommendation-of-the-investor-as-owner-subcommittee-on-esg-disclosure.pdf

Sen, S., Bhattacharya, C. B., & Korschun, D. (2006). The role of corporate social responsibility in strengthening multiple stakeholder relationships: A field experiment. *Journal of the Academy of Marketing science*, *34*(2), 158–166. https://doi.org/10.1177/0092070305284978

Siano, A., Vollero, A., Conte, F., & Amabile, S. (2017). "More than words": Expanding the taxonomy of greenwashing after the Volkswagen scandal. *Journal of Business Research*, *71*, 27–37. https://doi.org/10.1016/j.jbusres.2016.11.002

Turban, D. B., & Greening, D. W. (1997). Corporate social performance and organizational attractiveness to prospective employees. *The Academy of Management Journal*, *40*(3), 658–672. https://doi.org/10.2307/257057

Udayasankar, K. (2008). Corporate social responsibility and firm size. *Journal of Business Ethics*, *83*(2), 167–175. https://www.jstor.org/stable/25482364

Uyar, A., Karaman, A. S., & Kilic, M. (2020). Is corporate social responsibility reporting a tool of signaling or greenwashing? Evidence from the worldwide logistics sector. *Journal of Cleaner Production*, *253*, 119997. https://doi.org/10.1016/j.jclepro.2020.119997

Velte, P. (2022). Meta-analyses on corporate social responsibility (CSR): A literature review. *Management Review Quarterly*, *72*(3), 627–675. https://doi.org/10.1007/s11301-021-00211-2

Walker, K., & Wan, F. (2012). The harm of symbolic actions and greenwashing: Corporate actions and communications on environmental performance and their financial implications. *Journal of Business Ethics*, *109*, 227–242. https://doi.org/10.1007/s10551-011-1122-4

Withisuphakorn, P., & Jiraporn, P. (2016). The effect of firm maturity on corporate social responsibility (CSR): Do older firms invest more in CSR? *Applied Economics Letters*, *23*(4), 298–301. https://doi.org/10.1080/13504851.2015.1071464

Yang, Z., Nguyen, T. T. H., Nguyen, H. N., Nguyen, T. T. N., & Cao, T. T. (2020). Greenwashing behaviours: Causes, taxonomy and consequences based on a systematic literature review. *Journal of Business Economics and Management*, *21*(5), 1486–1507. https://doi.org/10.3846/jbem.2020.13225

Zerbib, O. D. (2019). The effect of pro-environmental preferences on bond prices: Evidence from green bonds. *Journal of Banking & Finance*, *98*, 39–60. https://doi.org/10.1016/j.jbankfin.2018.10.012

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APPENDIX A

See Table A1.

TABLE A1 Data definitions.

Acronym	Definition	Description	Source
Panel A: Firm/industry characteristics			
Env. Viol.	*Dum* = 1/0	For years 2008–2019, a firm/year instance is assigned *Dum* = 1 if it appears on the MSCI ESG KLD Stats file with a Negative Environment Performance Indicator regarding Hazardous Waste (Env-Con-A), *or* Regulatory Compliance (Env-Con-B), *or* Ozone Depleting Chemicals (Env-Con-C) *or* Toxic Emissions and Waste (Env-Con-D). For years 2020–2022, a firm/year instance is assigned *Dum* = 1 if it reports nonzero Environmental Fines (ENERDP103) in LSEG (Refinitiv)	KLD/ MSCI/LSEG
Size	*Price* × *N. of Shrs. Outstand.*	Market capitalization	CRSP
Age	Age of the firm	Period between the year of incorporation and the year of measurement	Compustat
Intangibility	1 − (*Tangible Assets*)/ (*Total Assets*)	% Of intangible assets of the firm	Compustat
ROE	($Net\ Income_t$)/ ($Equity_{t-1}$)	Return on equity (profitability)	Compustat
M/B	(*Market Cap*)/(*BV of Equity*)	Market-to-book ratio	Compustat/ CRSP
D/E	(*LT Debt*)/(*BV of Equity*)	Leverage ratio	Compustat
Industry D.	*Dum* = 1/0	A firm is assigned *Dum* = 1 (0) if is classified (not classified) in a specific industry according to the Kenneth French 12-industry classification.	Compustat
Foreign income	*Net Income*	*Net Income* attributed to international operations	Compustat
Foreign tax	*Tax Expensed*	Amount of Tax paid outside the country of incorporation	Compustat
International	*Dum* = 1/0	A firm is assigned *Dum* = 1 (0) for nonmissing (missing) Foreign Income of Foreign Tax	Compustat
Domestic	*Dum* = 1/0	A firm is assigned *Dum* = 1 if it is not International, 0 otherwise	Compustat.
NDir	*N. of Directors*	Number of Directors comprising the Board	Compustat

(Continues)

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TABLE A1 (Continued)

Acronym	Definition	Description	Source
%WomD	*% Women Directors*	Number of Female Directors as a % of *N. of Directors*	KLD/MSCI/ EDGAR
NDOut	*N. of Outside Directors*	Number of Directors outside the Firm	KLD/MSCI/ EDGAR
DispD	*Dum* = 1/0	A Firm is assigned *Dum* = 1 (0) for High (Low) Ownership Dispersion if its ownership structure is classified as "Institutions Dominant" or "Indexed Stock" or "Mixed Ownership" or "Widely Held" ("Principal Shareholder" or "Founder Firm" or "Family Firm" or "Controlling Shareholder")	KLD/MSCI/ EDGAR
Panel B: Natural-language processing variables			
N-pages	Number of pages	The page count in the CSR report (Size of the Report)	CSR Reports
N-words	Number of words	The number of words contained in the CSR report	CSR reports
Env-Sc	Environmental score	The Score quantifies the degree to which environmental content appears in the text analyzed	CSR reports
Read-Sc	Readability score	The Readability Score quantifies how difficult it is to read or understand the text analyzed	CSR reports
Pos-Sc	Positiveness score	The Positiveness Score quantifies the degree to which the text analyzed is positive, negative or neutral (sentiment analysis)	CSR reports
Env × *Pos*	*Env-Sc* × *Pos-Sc*	Interactive term between Environmental and Positiveness Scores	CSR reports
Env × *Read*	*Env-Sc* × *Read-Sc*	Interactive term between Environmental and Readability Scores	CSR reports

Abbreviations: CSR, corporate-social responsibility; CRSP, center for research in security prices; EDGAR, electronic data gathering, analysis, and retrieval; ESG, environmental, social, and governance; LSEG, London stock exchange group.

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